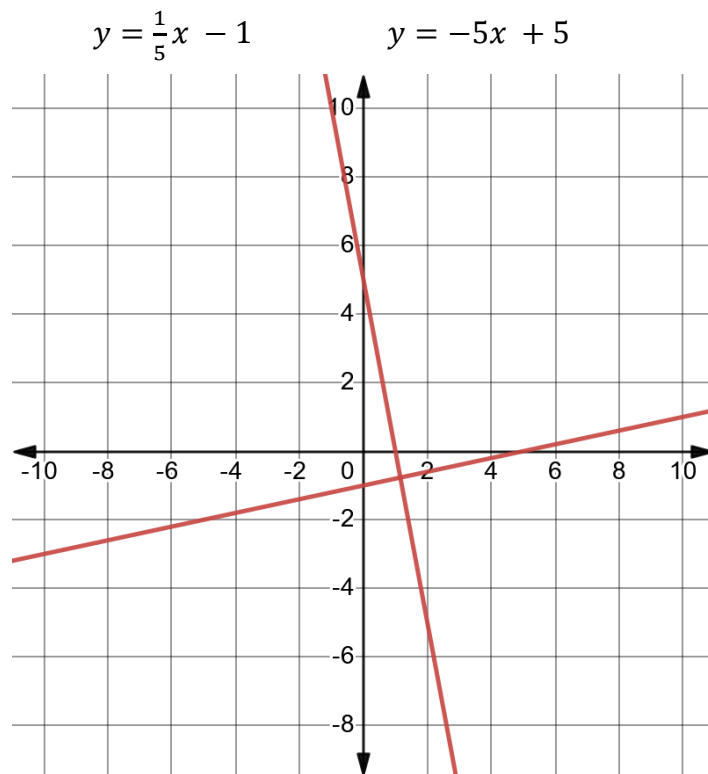


Geometry
Unit 13
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

1. Graph the equations on the coordinate plane.



2. The equations $y = \frac{1}{5}x - 1$ and $y = -5x + 5$ resulted in _____ lines.

☐ parallel
☒ perpendicular
☐ vertical

3. The slopes of the equations are _____

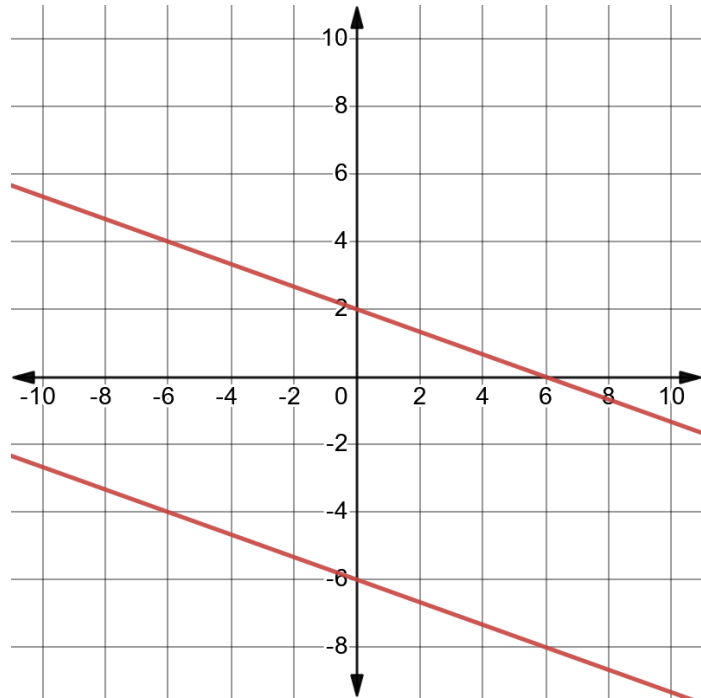
☐ equal.
☐ undefined.
☒ opposite reciprocals.

4. Rewrite the equation $x + 3y = -18$ in slope intercept form and graph the equation using slope-intercept form.

$$x + 3y = -18$$

$$3y = -x - 18$$

$$y = -\frac{1}{3}x - 6$$



5. Determine if $x + 3y = -18$ is parallel, perpendicular, or neither to the graph of the equation $y = -\frac{1}{3}x + 2$.

Parallel. The slopes are the same.

6. Graph the equation $y = -\frac{1}{3}x + 2$ on the graph to verify your answer to 5.

See graph.

7. Which equation represents a line which is perpendicular to the line $y = -\frac{8}{3}x - 1$?

A. $8x + 3y = 15$

B. $8y - 3x = 48$

C. $-3x - 8y = -56$

D. $8x - 3y = -24$

$$8y - 3x = 48$$

$$8y = 3x + 48$$

$$y = \frac{3}{8}x + 6$$

8. Which equation represents a line which is parallel to the line $3x + y = -40$?

A. $y = -3x + 3$

B. $y = 3x - 5$

C. $y = -\frac{1}{3}x + 8$

D. $y = \frac{1}{3}x - 2$

$$3x + y = -40$$

$$y = -3x - 40$$